

SUMMER TERM 2018

ECON3029: ADVANCED MACROECONOMICS SOLUTION

TIME ALLOWANCE: 2 hours

All questions should be answered.

Each part (A, B and C) carries 1/3 of the total mark. Questions B.1 and B.2 have the same weight.

PART A : A Model of Fertility and Growth

Consider an overlapping generation environment in discrete time. Economic agents live for two periods, childhood and adulthood. Population of adults is N_t . They do not take any decisions in childhood. Adults decide how much to consume of the unique good produced in the economy, and how much of their time they allocate to work and to child-rearing. Each adult has one unit of time. A child needs $0 < \phi < 1$ units of time to be raised, and the number of children per adult is n_t . For simplicity, we assume that this number does not need to be an integer. The time allocated to work is ℓ_t . Therefore, the time constraint of an adult is given by

$$\ell_t + \phi n_t = 1,$$

and the dynamics of the adult population is

$$N_{t+1} = n_t N_t.$$

Adults consume c_t . The consumption good is the numéraire, and the wage is denoted w_t . Adults preferences are given by

$$u(c_t, n_t) = c_t + \gamma \log n_t.$$

with $\gamma > 0$. The budget constraint of an adult is

$$c_t = w_t \ell_t.$$

- A.1 Write the utility maximisation problem of an adult. Substitute out ℓ_t and c_t for n_t using the constraints and derive the optimal fertility decision n_t . Comment on the obtained demand for children function.

The maximisation problem writes:

$$\begin{aligned} & \max_{c_t, n_t} c_t + \gamma \log n_t \\ \text{s.t. } & \begin{cases} c_t = w_t \ell_t \\ 1 = \ell_t + \phi n_t \end{cases} \end{aligned}$$

The constraints imply $c_t = w_t(1 - \phi n_t)$ so that the maximisation problem is :

$$\max_{n_t} w_t(1 - \phi n_t) + \gamma \log n_t.$$

The FOC is $w_t\phi - \frac{\gamma}{\phi n_t}$, so that the demand for children is

$$n_t = \frac{\gamma}{\phi w_t}.$$

On the production side, we assume that the production function is given by

$$C_t = \theta_t L_t,$$

where $\theta_t > 0$ is the level of productivity and L_t is total employment.

- A.2 Write the profit maximisation problem of a competitive representative firm, and explain why the wage w_t must be equal to θ_t in equilibrium.

The profit maximisation problem writes

$$\begin{aligned} & \max_{C_t, L_t} C_t - w_t L_t \\ \text{s.t. } & C_t = \theta_t L_t. \end{aligned}$$

This can be rewritten as

$$\max_{L_t} (\theta_t - w_t) L_t.$$

The objective function is not strictly quasi-concave (it is linear), and therefore one cannot use FOC. The solution to this maximisation problem is

- (a) if $\theta_t > w_t$, then the representative firm is willing to let L_t tend to $+\infty$
- (b) if $\theta_t < w_t$, then the representative firm is not willing to hire and the optimal solution is $L_t = 0$

(c) if $\theta_t = w_t$, then the representative firm is indifferent and can hire any level of employees L_t .

We will have an equilibrium that will be interior for $w_t = \theta_t$ (see below)

We now study the equilibrium and the dynamics of this economy.

A.3 What are the variables that are either exogenous or predetermined in period t ?

θ_t is exogenous and N_t is predetermined.

A.4 What is the equilibrium value of n_t ? Of income per adult $y_t = \frac{C_t}{N_t}$?

In equilibrium, labor supply equals labor demand. Labor supply is strictly positive and finite for strictly positive wage. The only possible equilibrium is therefore that labor demand is also positive and finite, which happens when $w_t = \theta_t$. We therefore have $n_t = \frac{\gamma}{\phi\theta_t}$, and $\ell_t = 1 - \phi n_t = \frac{\theta_t - 1}{\theta_t}$. Total hours worked are $L_t = N_t \ell_t = \frac{\theta_t - 1}{\theta_t} N_t$.

Income per adult is $y_t = \frac{C_t}{N_t} = \frac{\theta_t L_t}{N_t} = \theta_t - \gamma$.

A.5 Find the difference equation $N_{t+1} = f(N_t)$ that determines adult population dynamics.

The law of motion of N is

$$N_{t+1} = N_t n_t = \frac{\gamma}{\phi\theta_t} N_t.$$

A.6 Assume $\theta_t = \theta \forall t$ (no technical progress). Draw f and the dynamics of N in the space (N_t, N_{t+1}) , for the two cases $\frac{\gamma}{\theta\phi} > 1$ and $\frac{\gamma}{\theta\phi} < 1$. In each of those two cases, what is the limit of N_t when $t \rightarrow +\infty$? Give an interpretation.

See Figure 1.

When $\frac{\gamma}{\theta\phi} > 1$, the time needed to raise a children (ϕ small) is relatively small, the utility derived from having children is relatively large (γ large) and productivity is relatively small (θ small), then each adult has more than one children, so that population is exploding. In the opposite case, each adult has less than one child, and population is shrinking to zero. Productivity affects negatively population growth because the cost of raising children is in units of time, and therefore in terms of foregone wage. When labor is more productive, the wage is higher, and so is the opportunity cost of raising children.

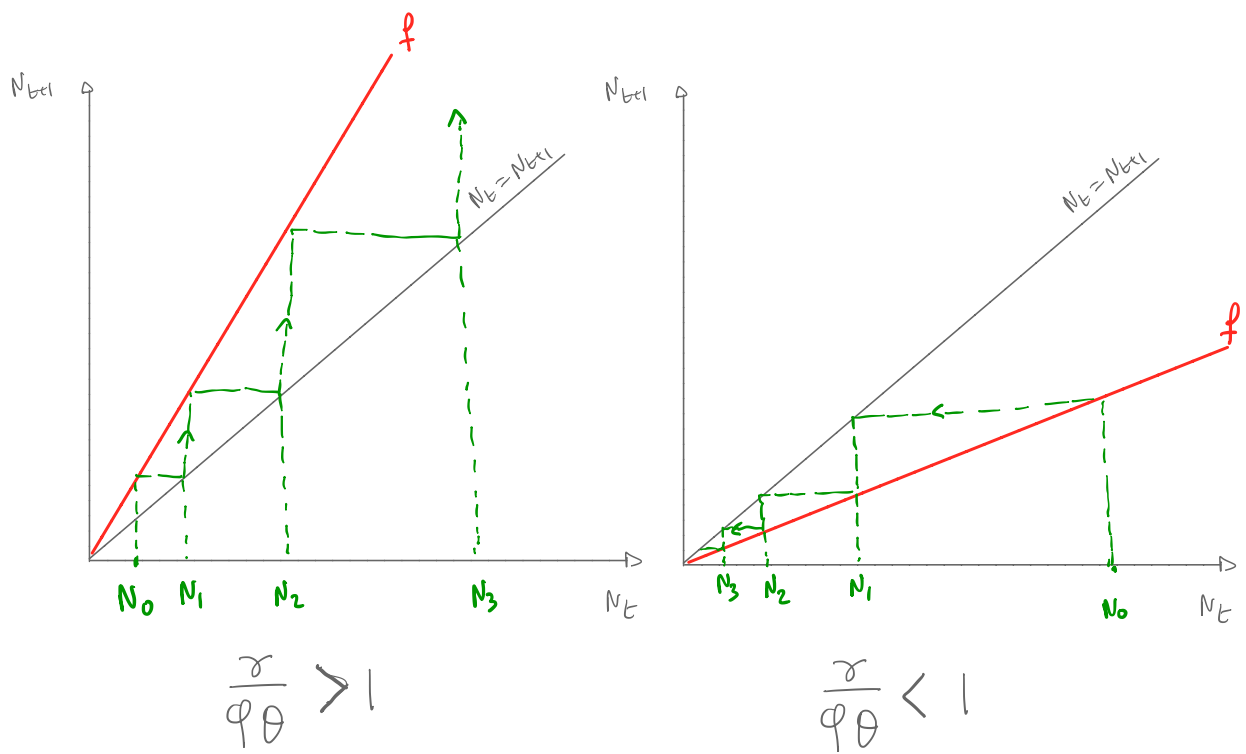
A.7 Assume now that there is some technological progress, so that $\theta_t = \theta_0 \rho^t$ with $\rho > 1$. Use the equilibrium expression for n_t to find the limit of n_t and N_t when $t \rightarrow +\infty$.

In this case,

$$n_t = \frac{\gamma}{\phi\theta_0} \rho^{-t},$$

so that the limit of n_t when t goes to infinity is zero.

Figure 1: Dynamics of Population



A.8 What are the economic forces driving the result of the previous question?

As I already explained in A.6., the cost of raising children is in units of time, and therefore in terms of foregone wage. When labor is more productive, the wage is higher, and so is the opportunity cost of raising children.

PART B : Questions

For each question, you should give a structured answer.

- B.1 PIKETTY's two "laws of capitalism" are $\sigma_k = r \times \frac{k}{y}$ (first "law") and $\frac{k}{y} = \frac{s}{g}$ (second "law"), where σ_k is the share of capital in total income, k is capital, y is income, s is the saving rate, r the interest rate and g the growth rate of income. Discuss those two laws and their implication when $r > g$, when g tends to zero?

- × The first "law" is

$$\sigma_k = r \times \frac{k}{y}.$$

It is not a law but a definition. rk represent capital income and y total income, so that *by definition* $\frac{rk}{y}$ is the share of capital income in total income.

- × σ_k is a measure of inequality because in most (all) of the countries, most of capital is concentrated in the hands of a small share of the population.
- × The second law is

$$\frac{k}{y} = \frac{s}{g}.$$

It states that the size of the capital stock, in relation to total income (often referred as the capital-income or capital-output ratio), is equal to the saving rate over the growth rate of the economy. The saving rate s determines the increase in the capital stock from one period to another, while the growth rate g determines the increase in income.

- × Using the two laws, we can obtain:

$$\sigma_k = s \frac{r}{g}.$$

The ratio $\frac{r}{g}$ matters for inequalities.

- × The case $r > g$ is discussed by PIKETTY because it implies that $\frac{k}{y}$ is large, and therefore inequalities are large. According to his studies, we are entering in a period in which r will be systematically larger than g .
- × The case g tends to zero is seen as particularly worrisome by PIKETTY. Because developed economies have entered in a period of very low growth (called secular stagnation by some economists), the share of capital in income is likely to be very large, hence inequalities are expected to rise.
- × Of course, an increase of inequalities is a concern if, from a *normative* point of view, one believes that inequalities are bad. We have not discussed normative questions in the lecture on inequalities.

- × The analysis of PIKETTY relies on two strong assumptions : (i) that s is constant and (ii) that r does not vary when $\frac{k}{y}$ varies. Those two assumptions are not reasonable for large variations of $\frac{k}{y}$.
- × Assumption that s is constant : high $\frac{k}{y}$ implies a lot of depreciation δk , so that very little *net output* is left for consumption is the saving rate applies to *net output*. It is unlikely then that s would stay constant. And if saving rate applies to *gross output*, the $\frac{k}{y}$ is bounded when g goes to zero.
- × Assumption that r is constant : when k/y increases, does r stays constant or varies? If r decrease a lot, then σ_k will not increase when k/y increase. With Cobb-Douglas production function, r decreases proportionally to $\frac{k}{y}$ so that σ_k is constant. If capital and labor are complement (which is likely to be the case in reality), then r decreases more than proportionally compared to $\frac{k}{y}$, so that σ_k decreases with k/y . Therefore, capital share of income is unlikely to increase (at least in the long run) if r increases.

B.2 What are the economic forces that link economic growth to natural resources and their dynamics?

- × Natural resources can be oil or coal in the ground, fish population in the sea, forests, the quality of the atmosphere, biodiversity, etc....
- × As we have discussed in class in the case of Easter Island, there is a dual relationship between natural resources and economic growth.
- × First, natural resources (say trees) are useful for economic growth. They are used to produce.
- × Second, economic growth is using natural resources, and therefore depleting their stock
- × The important question is how fast the economy is depleting the stock of natural resources in comparison to the natural regeneration of the stock.
 - In the case of resources that are in finite supply (say oil, the economy needs to find a way (technological innovation) to be able to grow without them once they will be exhausted, because we cannot produce oil (or very inefficiently).
 - In the case of renewable resources (fish, forest), one needs to find a *sustainable* path, along which the stock regenerates enough to compensate for the use of the resource by the economy. This is what we call *sustainable development*.
 - In the case of Easter Island (and according to one particular theory of what happened, as developed by JARRED DIAMOND), one would have needed to cut trees at a pace that would have allowed to maintain the forest size constant.
- × Often, there is *irreversibility* in natural resource dynamics. For example, once the last tree is cut, there is no more natural regeneration of the forest.

PART C : Discussion

Below is an article posted on the website *The Market Mogul* on August 18, 2016 and entitled “The UK’s Liquidity Trap”. Read carefully this article before answering the questions below.

C.1 Write a 10-line summary of that article.

- Since the 2007-08 financial crisis, the UK economy is in a liquidity trap: interest rates are low (0.25%), bond yields are low and there is under-utilization of capacities.
 - The bond market is particularly depressed. Yields on government bonds are now negative (-0.1%). This has a depressing effect of pensions.
 - Monetary policy is quite ineffective, despite a huge stimulus since 2009. The gains from the reduction in interest rates might have been offset by contractionary fiscal policy measures.
 - The new government, despite some announces, has not yet implemented an active fiscal policy that would reduce taxes and increase government spending.
- The UK seems to head to the situation in which Japan has been since the 1990’s.
- One positive note is that the UK equities market is quite strong.
 - Where do we go? Monetary policy cannot be more loose. It is difficult time for the UK savers and pensioners. One must hope that the latest monetary stimulus will keep the economy moving.

C.2 What are the characteristics of a liquidity trap according to the author? Do you know other historical episodes of a liquidity trap?

The characteristics of a liquidity trap according to the author are low-interest rates, low bond yields and an economy operating below capacity (meaning that there is also unemployment). As we saw in class, this is a situation that happened in the past. Examples are the Japanese stagnation of the 1990’s and the Great Recession of the 1930’s in the U.S.A.

C.3 Why would inflation be bad for investors? Is that necessarily bad for the economy as a whole in such a situation? Why?

By investors, the author means investors on financial markets. Financial investors care about the real return on their investment, which is defined as the nominal return minus expected inflation. If inflation increases (while the nominal interest rate stays stuck at zero), then real return decreases, which these investors do not like. But an increase in expected inflation might be good for the economy because decreasing the real interest rate is good for households that borrow to buy houses and durable goods (cars, tv sets, etc..). It is also good for entrepreneurs that borrow to launch new firms. For those economic agents, an increase in inflation decrease the cost of spending, creating firms and hiring, which is likely to push up demand and employment.

C.4 The author writes “ *It could be argued that now is the time to start taking decisive action by reducing taxes and increasing government spending to move the economy out of the stagnant position it finds itself in.*” Does it make sense?

Yes, it does make sense in an IS-LM type of model or in the model we have seen in class. In the liquidity trap, monetary policy is inefficient because the nominal interest rate is at the zero lower bound. Fiscal policy is the efficient alternative in that situation.

C.5 Find in this article an idea or a statement with which one could disagree, and explain why.

Here there is not a single correct answer. I would I said the following (but many other answers are possible if well argued)

The author writes “*One must hope that Carney’s latest monetary stimulus will keep the economy moving, otherwise the BoE may be forced into taking some potentially inflationary fiscal measures.*” As we have seen in class, the problem in the liquidity trap is that monetary policy cannot decrease anymore the real interest rate because the nominal interest rate is at the zero lower bound. In that case, it is not a problem that the BoE takes potentially inflationary measures. Indeed, increasing inflation expectation is one way of exiting the liquidity trap. This is what KRUGMAN calls “committing to be irresponsible”.

THE UK'S LIQUIDITY TRAP, by Sam Lerche , *The Market Mogul*, August 18, 2016

Britain is in the midst of an economic slowdown, which started after the 2007-2008 financial crisis and has resurfaced following the UK's vote to leave the EU. The UK has just reduced its interest rates to a new record low of 0.25% in the wake of Brexit, having previously been 0.5% since March 2009. These low-interest rates coupled with tiny bond yields and an economy operating below capacity suggest that all the characteristics of a liquidity trap are being fulfilled. The US, Japan and much of Europe are also experiencing similar conditions, but there is no clear policy to "escape" from the trap, although significant monetary stimulus is the way most policy makers are going.

The Bond Market

One major area of concern for the UK is the increasing price of government bonds and consequently the diminishing yields on them. The government announced plans for a new £60bn round of quantitative easing together with the cut in interest rates. The idea behind this is to encourage the sellers of government bonds to invest the money back into the private sector, which supports growth. The buyback of government bonds will only push up prices, which decreases yields.

Last week, yields on gilts maturing in 2019 and 2020 were at -0.1%, meaning that buying these bonds will get a negative return. In real terms, this problem for investors is multiplied with the expected rise in inflation following the sharp devaluation of the pound and rising commodity prices. One of the main consequences of these tiny to negative yields in the bond market is the effect this has on individual pensions. Ultimately, lower yields result in lower returns for their savings as pension funds buy government bonds as liability for their annuities.

The Monetary Policies

Another major implication of the liquidity trap is that it makes the main macroeconomic tool, monetary policy, largely ineffective. Clearly, the huge monetary stimulus, which started with the drop in interest rates to 0.5% in 2009, hasn't had the desired outcome on economic performance. Perhaps a reason for this is the contractionary fiscal policy measures introduced by George Osborne in 2010, which may have offset gains achieved from the drop in interest rates.

The new government under Theresa May now finds itself in a state of limbo by declaring its aim to reduce austerity measures, yet not committing to an active fiscal policy. It could be argued that now is the time to start taking decisive action by reducing taxes and increasing government spending to move the economy out of the stagnant position it finds itself in.

Furthermore, a show of boldness and courage by the Bank of England (BoE) could help exert confidence back to the consumer. However, this shift in stance by the BOE could also put even more inflationary pressure on the economy, which they are evidently trying to avoid.

Is Japan Where The UK Is Heading?

Unfortunately, the UK is not alone in this situation. The world's third- biggest economy, Japan, first experienced conditions associated with a liquidity trap in the 1990s and recent data would suggest that they still haven't escaped the bottleneck. One economic argument is that when rates go negative like they have in Japan, they reduce the velocity of money through the economy as people take their money out the banks and hoard it at home or in other units of value such as commodities. This has a deflationary impact on the economy as each yen generates less economic activity. Therefore policymakers must put in more and more money to try and generate growth. One has seen another example of this with Abe's most recent \$258bn stimulus package. Furthermore, Japan is experiencing flatline inflation, which makes real interest rates more attractive, thus keeping the value of the yen strong and this, in turn, has an adverse impact on the country's trade balance. Although the UK is in a much stronger position than Japan, it shows the damaging long-term consequences of a fledgeling economy and keeping interest rates too low for too long.

UK Equities ? The Resurgence

The UK equities market has surged to almost record highs in the wake of these challenging economic conditions. The severely weakened pound along with comparatively poor yields in the bond market have been the main drivers of this resurgence, and J.P. Morgan remains bullish that stock prices will remain high despite negative economic data. In this sense, the damaging impact of the liquidity trap has been partly recuperated by gains in UK equities but how much this affects the ordinary working man is questionable, to say the least. Nonetheless, it does at least give investors an opportunity for short-term gains.

Conclusion

It is a worrying time for the UK. It is clear that monetary policy cannot be loosened much further with Mark Carney, Governor of the BoE, already saying that the UK will not follow some of its European counterparts in introducing negative interest rates. Clearly, the UK's difficult position will have an effect on individuals whether they are pensioners or for ordinary savers. The expected rise in inflation caused by the weakening pound and rising commodity prices will only accelerate the decline in people's real savings. One must hope that Carney's latest monetary stimulus will keep the economy moving, otherwise the BoE may be forced into taking some potentially inflationary fiscal measures.